

LO 3.9.A Inverse Trigonometric Functions — Key

Key Vocabulary

- **arcsine** — the inverse of the restricted sine function; outputs an angle in $[-\frac{\pi}{2}, \frac{\pi}{2}]$
- **arccosine** — the inverse of the restricted cosine function; outputs an angle in $[0, \pi]$
- **arctangent** — the inverse of the restricted tangent function; outputs an angle in $(-\frac{\pi}{2}, \frac{\pi}{2})$
- **restricted domain** — a subset of the natural domain chosen so a function is one-to-one
- **principal value** — the unique output angle in the restricted range of an inverse trig function

Section 1 — Why Do We Need Restricted Domains?

The table below shows selected values of $f(x) = \sin x$.

x	$-\frac{\pi}{2}$	$-\frac{\pi}{6}$	0	$\frac{\pi}{6}$	$\frac{\pi}{2}$	$\frac{5\pi}{6}$	π
$\sin x$	-1	$-\frac{1}{2}$	0	$\frac{1}{2}$	1	$\frac{1}{2}$	0

$\sin x$ is **NOT one-to-one**.

Because the sine function is periodic, it **fails** the horizontal line test, so it **does not** have an inverse over its full domain.

Section 2 — Restricted Domains and the Inverse Trig Functions

Function	Restricted Domain	Range	Inverse Name	Notation
Sine	$[-\frac{\pi}{2}, \frac{\pi}{2}]$	$[-1, 1]$	Arcsine	$\arcsin x$
Cosine	$[0, \pi]$	$[-1, 1]$	Arccosine	$\arccos x$
Tangent	$(-\frac{\pi}{2}, \frac{\pi}{2})$	$(-\infty, \infty)$	Arctangent	$\arctan x$

The output of an inverse trig function is an **angle** measure (the principal value).

Section 3 — Evaluating Inverse Trig Functions

Expression	Think:	Answer
$\arcsin\left(\frac{1}{2}\right)$	Which $\theta \in [-\frac{\pi}{2}, \frac{\pi}{2}]$ has $\sin \theta = \frac{1}{2}$?	$\frac{\pi}{6}$
$\arccos(0)$	Which $\theta \in [0, \pi]$ has $\cos \theta = 0$?	$\frac{\pi}{2}$
$\arctan(1)$	Which $\theta \in (-\frac{\pi}{2}, \frac{\pi}{2})$ has $\tan \theta = 1$?	$\frac{\pi}{4}$
$\arcsin\left(-\frac{\sqrt{2}}{2}\right)$	Which $\theta \in [-\frac{\pi}{2}, \frac{\pi}{2}]$ has $\sin \theta = -\frac{\sqrt{2}}{2}$?	$-\frac{\pi}{4}$
$\arccos(-1)$	Which $\theta \in [0, \pi]$ has $\cos \theta = -1$?	π

Guided Practice

Practice: Evaluate — Key

$$1. \arcsin(1) = \frac{\pi}{2}$$

$$4. \sin^{-1}\left(-\frac{1}{2}\right) = -\frac{\pi}{6}$$

$$2. \arccos\left(\frac{\sqrt{3}}{2}\right) = \frac{\pi}{6}$$

$$5. \cos^{-1}\left(-\frac{\sqrt{2}}{2}\right) = \frac{3\pi}{4}$$

$$3. \arctan(-1) = -\frac{\pi}{4}$$

$$6. \tan^{-1}(0) = 0$$